

Soumil Chugh

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GitHub

🌐 <https://soumilchugh.github.io/>

Summary

AI/ML engineer with 8+ years of experience building production AI systems across LLM agents, multimodal AI, computer vision, and edge ML. Experienced in architecting agentic workflows, retrieval systems, and research-to-product ML initiatives. Inventor on 15+ patents with 7 peer-reviewed publications.

Technical Skills

AI/ML	LLMs, Agentic AI, Multi-Agent Systems, RAG, Graph-RAG, Multimodal AI, VLMs, Computer Vision, Fine-tuning, PEFT/LoRA, Model Evaluation
Frameworks	PyTorch, Hugging Face Transformers, LangChain/LangGraph, OpenAI API, TensorFlow, OpenCV
AI Infrastructure	Vector Databases, Embedding Systems, Model Serving, LLM Evaluation Pipelines, Prompt Engineering, MLOps
Cloud & Systems Programming	AWS, Docker, CI/CD, Distributed Systems, REST APIs, WebSockets, Git
	Python, C++, TypeScript, Java, SQL

Employment History

Sep 2025 – ... **Senior Software Engineer - AI Agent Systems**, Kyoso, Toronto, Canada

- Architecting an AI co-creation platform enabling advertisers to generate and modify brand-compliant creatives through autonomous agent workflows.
- Designed multimodal reasoning systems combining VLMs, structured design representations, and tool execution for understanding and editing visual content.
- Developed scalable knowledge infrastructure integrating brand guidelines, design assets, and historical context to provide persistent memory for AI systems.
- Built evaluation pipelines using LLM-as-judge frameworks to measure creative quality, brand alignment, and agent execution reliability.
- Led architecture discussions and established engineering patterns for building reliable AI agents, including orchestration flows, tool interfaces, memory systems, and evaluation strategies.

Employment History (continued)

- Aug 2020 – Sep 2025  **Senior AI/ML Research Engineer**, Huawei Technologies, Toronto, Canada.
- Led technical direction for AI assistant initiatives, designing LLM agent architectures for intelligent photo editing, creative assistance, and browser automation.
 - Architected a deep research agent combining multi-step planning, RAG, web retrieval, synthesis, and automated visualization for analytical report generation.
 - Developed a browser automation agent capable of observing workflows, reasoning over interfaces, and executing multi-step tasks using LLM planning and tool execution.
 - Owned multimodal interaction frameworks combining gesture recognition models and LLM function calling, achieving 90% precision and recall.
 - Adapted foundation models using LoRA/PEFT pipelines for domain-specific creative generation including stickers and color palettes.
 - Led multi-year computer vision initiatives from research to deployment, improving gaze estimation accuracy by 25% and reducing inference latency by 50%.
- Aug 2020 – May 2024  **Machine Learning Consultant (Part-Time)**, General Prognostics (GPx), Boston, USA.
- Architected smartwatch-based ML pipelines processing biometric signals (HR, HRV, activity, sleep) for predictive healthcare analytics.
 - Developed ensemble ML models for early detection of patient health changes using multimodal time-series data.
- Sep 2015 – July 2018  **Senior Software Engineer**, Jana Care, Bengaluru, India.
- Led software development for an FDA-cleared smartphone-connected diagnostic robotics platform integrating Android, embedded systems, BLE, and computer vision.

Education

- Aug 2018 – Aug 2020  **MASc., University of Toronto, Canada** in Computer Engineering.
Thesis title: *Eye Tracking System for a Virtual Reality Headset*.
- Aug 2011- May 2015  **BE., Punjab University, India** in Electronics and Communication.
Thesis title: *Non-invasive hemoglobin monitoring device*.

Patents & Publications

- Inventor on 15+ granted/filed patents across AI agents, computer vision, gaze estimation, and human-computer interaction.
- Author of 7 peer-reviewed publications in computer vision, medical AI, and human-computer interaction.